Pre-registration T7: the flywheel inside the Octavian Catch

Written by Claude (Opus 5.5) at Pedro Garcia's request. Registered before any code for T7 was run.

24 September 2026

Object. The octonion table and the 7 closed Octavian unit sets of T6 (same code, same integers scaled by 2, no floats). The table's 21 sign-exact symmetries include elements of order 3 (Fano relabelings of the form $x \mapsto 2x$). A *flywheel triple* is three closed sets $A \to B \to C \to A$ cycled by one order-3 symmetry: three kernels a third of a turn apart.

Operationalisation (Claude's, declared). Pedro's posit: three kernels (compute, pace, memstore) turning together close as a flywheel, and the outside export explains a missing 30°. Read here as: the *export* of a triple is the set of cross products uv, $u \in A$, $v \in B$ ($240^2 = 57{,}600$ products); a product is *caught* iff it lies in $A \cup B \cup C$; 30° is one twelfth of a turn, read as a caught-fraction deficit of exactly 1/12 (4,800 products).

Hypotheses and kills.

H1 Structure. Each order-3 symmetry acts on the 7 closed sets as one fixed set plus two 3-cycles. *Kill:* any other cycle type.

H2 (Claude) The flywheel closes: every cross product of a triple is caught (0 escape). *Kill:* any escape.

H3 (Pedro, as operationalised) Exactly 1/12 of cross products escape the triple (4,800 of 57,600). *Kill:* any other count.

H2 and H3 compete; at most one can hold.

E1 Exploratory: where escaping products land (the fixed set, the other triple, another labeling, or a quarter-integer rung).

Outcomes are appended below this line and never edited into the text above.

Outcomes (appended after the run; registered source SHA-256 `214f247c...e84d`).

H1 **Holds.** All 14 order-3 symmetries act on the 7 closed sets as $1 + 3 + 3$: one fixed hub, two 3-cycles.

H2 **Killed.** Triples do not close: 35,840 or 36,864 of 57,600 cross products escape.

H3 **Killed.** The escape count is not 4,800 in any triple checked.

E1 Every escape lands either on the quarter-integer rung or inside another closed set; none lands in an open labeling or in no set.

Coverage disclosed. 6 of the 84 ordered triples were run (time budget). Both kills trigger on every triple run, so coverage does not change a verdict; the escape census is for those 6. Two triple types appeared, 3 each. **Post-hoc, not registered.** Caught counts are 21,760 and $20{,}736 = 12^4$. Recorded only as candidates for a later registration.